Public

AccelQ3D: Extending FINN for FPGA acceleration of 3D CNNs

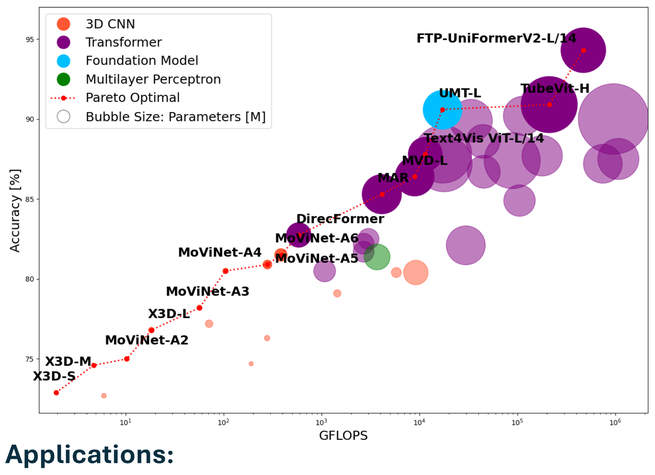
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Motivation

Why 3D CNNs? They capture both spatial and temporal information, making them ideal for understanding motion and change over time. The graph below shows that 3D CNNs offer a favorable trade-off between accuracy and computational cost (parameters and FLOPS) compared to architectures such as Transformers, which makes them suitable for deployment on memory and power-constrained edge devices.

Why deploy 3D CNNs on FPGAs?

3D CNNs are particularly useful for real-time edge applications where throughput, latency and power consumption are critical.
Custom FPGA accelerators for 3D CNNs can achieve an optimal balance between high throughput, low latency and low power consumption, outperforming CPUs and GPUs.



Applications:

Autonomous Robots | Medical Imaging | Advanced driver-assistance systems

Contribution

Brevitas + AccelQ3D Extension

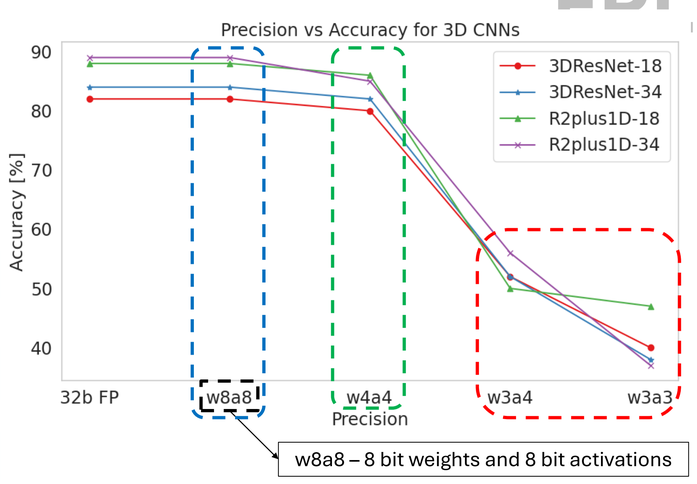
Quantization of 3D CNNs can dramatically reduce the compute and memory requirements of FPGA implementations. Low-bit (< 8 bit) quantization of 3D CNNs has not been previously explored. Brevitas was extended to handle quantization-aware training of 3D CNNs.

3D CNNs introduce an additional dimension -> 5D tensor shapes: [N,D,H,W,C].
Brevitas was extended with modules corresponding to different layers, such as QuantConv3d and QuantMaxPool3d, and the 5D data layout. Extending FINN is work in progress but it currently recognizes these new layers and data layout.

Test 3D CNNs: **3DResNet** and **R2plus1D** with 18 and 34 layers

→ Pre-trained on the Kinetics-700* dataset → Trained on the UCF101* dataset

*Action recognition datasets; UCF101 is a common benchmark in majority of published work



All networks follow the same trend: 8-bit quantization does not impact the accuracy.

When quantizing to 4 bits for both weights and activations we observe a decline of 2-4 percentage points. **W4A4 represents the best quantization-accuracy trade-off.**

Accuracy decreases for W3A4 and W3A3 models. Going under 4 bits for weights significantly impacts the ability of the network to learn patterns and maintain pre-trained information.

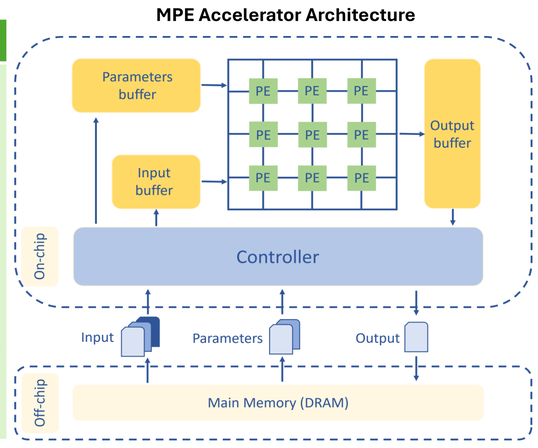
Key ideas

All current FPGA accelerators for 3D CNNs use a Matrix of Processing Elements (MPE) architecture, a general-purpose structure designed to accommodate a range of network types. The DataFlow (DF) architectures generated through **FINN** can outperform MPEs by being fully customized to a target 3D CNN.

Advantages

Reusable for different DNN topologies without regenerating the hardware; more accessible for non-experts.
Design exploration and optimization is done only once.

MPE Accelerator Architecture



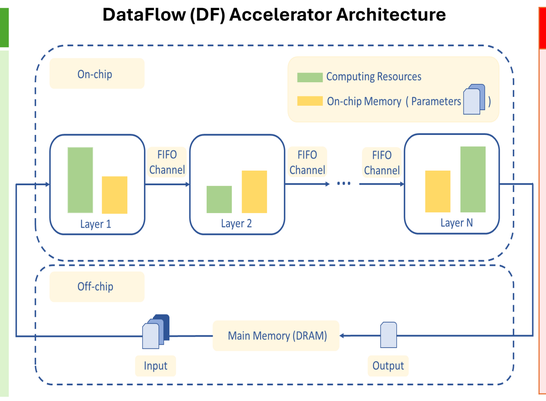
Disadvantages

Fixed precision.
Intensive data movement between on-chip and off-chip memory.
May involve communication with a CPU.
Limited customization for different types of DNNs.

Advantages

Leverages customization for each target DNN which can provide better performance and better power efficiency.
Adjustable parallelism.
Variable precision.
Minimal communication with off-chip memory.

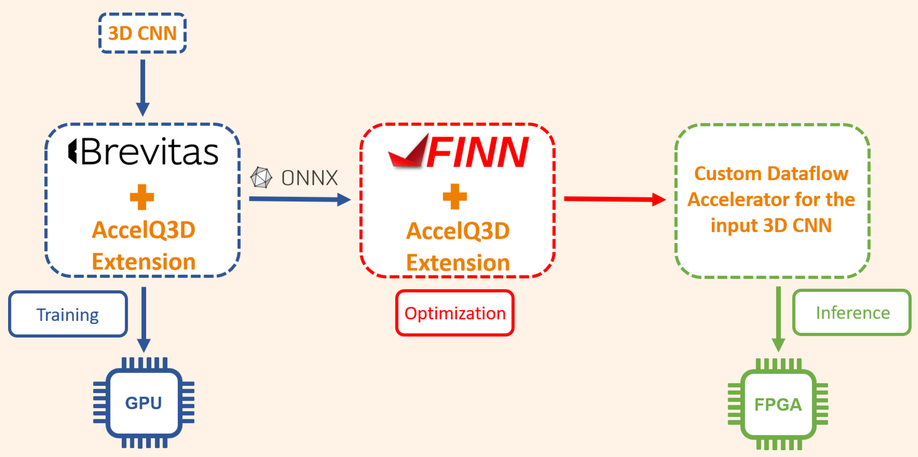
DataFlow (DF) Accelerator Architecture



Disadvantages

Custom design must be regenerated for different networks or parameters; hardware compilation takes a significant amount of time.
Requires a certain level of expertise to optimize the design and maximize performance.

AccelQ3D extends **FINN** to build custom dataflow accelerators for 3D CNNs. The workflow extends Brevitas to support quantization-aware training of 3D CNNs, including layers, operators and data representations. The quantized models are exported to an extended FINN flow capable of recognizing the same layers, operators and data layouts, translating the layers to HLS code and generating efficient FPGA accelerators.



Future Work

Brevitas + AccelQ3D Extension

- Further exploration on the effect of low-bit quantization on other 3D CNNs architectures.
- Evaluate more mixed-precision configurations to find the best balance between number of bits and accuracy.

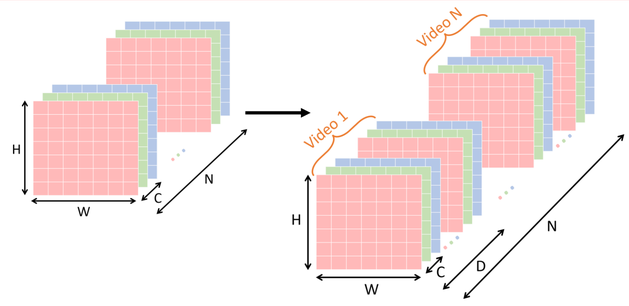
FINN + AccelQ3D Extension

- Complete **AccelQ3D** extension of **FINN** to support 3D layers and data representation (5D Tensor shape).
- Design the HLS module for 3D convolution and compare it with other efficient techniques such as the Winograd transformation, which reduces the number of multiplications at the cost of more additions.
- Optimize the designs and evaluate them in terms of throughput, latency, resource utilization and power consumption.
- Complete accelerator build flows for different 3D CNNs, such as 3DResNet and R2plus1D.

Discussion

Why FINN?

- Custom dataflow architectures for a target 3D CNN can achieve the best performance-power trade-off.
- FINN's modularity makes it accessible for researchers to extend the framework with new operators and transformations.



2D CNN input representation
4D Tensor shape : [N, C, H, W] or [N, H, W, C]

3D CNN input representation
5D Tensor shape : [N, C, D, H, W] or [N, D, H, W, C]

Challenges

- Previous experiments with a 2D CNN, ResNet50, quantized to W4A5, showed that weights for the last fully-connected layer needed to be stored off-chip. With the additional dimension, 3DResNet (and other 3D CNNs) are expected to introduce greater storage challenges.
- A highly parallel, fully unrolled dataflow accelerator of a W4A4 3D CNN may not fit on one FPGA due to its increased resource requirements. Splitting the kernels across devices can also be challenging. Reducing the level of parallelism can lower resource usage, but it may also reduce performance to a level comparable to an MPE.

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